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Inventor(s)

KONDO; Yusuke et al.

VEHICLE DOOR HANDLE DEVICE

Abstract

A vehicle door handle device includes a guide member and a latch release lever. The guide member includes a concave portion into which a user can insert his or her finger from a vehicle outer side. The latch release lever switches a door lock device from a latching state to an unlatching state by moving from an initial position to an actuation position relative to the guide member. An inner peripheral surface of the concave portion includes an inner peripheral surface formation portion located on a vehicle inner side of a vehicle-outer-side surface of the vehicle door and located at an end portion of the concave portion on one side in a front-rear direction, and faces to a side substantially opposite to the one side in the front-rear direction. The latch release lever forms at least a part of the inner peripheral surface formation portion.

Inventors: KONDO; Yusuke (Kariya-shi, JP), TAKENAKA; Kazuki (Kariya-shi, JP), NAGASAKI; Tetsuya (Nagoya-shi, JP)

Applicant: AISIN CORPORATION (Kariya, JP); TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

Family ID: 1000008419990

Assignee: AISIN CORPORATION (Kariya, JP); TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Japanese Patent Application 2024-026560, filed on Feb. 26, 2024, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] This disclosure generally relates to a vehicle door handle device.

BACKGROUND DISCUSSION

[0003] The vehicle door handle device disclosed in WO2019/120599A1 (Reference 1) includes an operation member (button: taster) rotatably attached to a frame, and a microswitch (mikroschalters) that is actuated by a movement of the operation member. A user can open a vehicle door by inserting his or her hand into a concave portion on the frame, and rotating the operation member and then actuating the microswitch.

[0004] Further, the vehicle door handle device disclosed in Reference 1 includes a latch release lever (emergency operation lever: notbetätigungshebels). The latch release lever is a member used for opening the vehicle door when the vehicle door cannot be opened by use of the operation member due to an abnormality or the like in a control device. The latch release lever is configured to be movable between an initial position (ruheposition) and an actuation position (betätigungsposition). A user can open the vehicle door by hooking his or her finger onto the latch release lever through the concave portion on the frame and pulling out the latch release lever. In principle, the latch release lever is used only for opening the vehicle door in an emergency, and is not used at a normal time.

[0005] In the vehicle door handle device disclosed in Reference 1, the initial position of the latch release lever is a position at which a vehicle-outer-side surface of the latch release lever becomes flush with the frame of the door handle device. In this configuration, the latch release lever is noticeable (particularly, a front end portion (edge) of the latch release lever is noticeable), and thus, it is feared that a user may mistakenly recognize the latch release lever as the operation member to be normally used for opening the vehicle door. The noticeable latch release lever impairs an aesthetic appearance of the door handle device.

[0006] A need thus exists for a vehicle door handle device, which is not susceptible to the drawback mentioned above.

SUMMARY

[0007] A vehicle door handle device according to this disclosure is attached to a vehicle door including a door lock device that is switchable between a latching state of not allowing the vehicle door to be opened and an unlatching state of allowing the vehicle door to be opened. The vehicle door handle device includes a casing and a latch release lever. The casing is open on a vehicle outer side, and includes a concave portion into which a user can insert his or her finger from a vehicle outer side. The latch release lever is attached to the casing in such a way as to be movable between an initial position and an actuation position. The latch release lever is linked to the door lock device in such a way as to switch the door lock device from the latching state to the unlatching state by moving from the initial position to the actuation position. An inner peripheral surface of the concave portion includes an inner peripheral surface formation portion that is located on a vehicle inner side of a vehicle-outer-side surface of the vehicle door and that is located at an end portion of

the concave portion on one side in a front-rear direction of the vehicle door. The inner peripheral surface formation portion faces to a side substantially opposite to the one side in the front-rear direction of the vehicle door. The latch release lever forms at least a part of the inner peripheral surface formation portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The foregoing and additional features and characteristics of this disclosure will become more apparent from the following detailed description considered with the reference to the accompanying drawings, wherein:

[0009] FIG. 1 is a schematical view illustrating a configuration of a vehicle door to which a vehicle door handle device according to an embodiment of this disclosure is applied;

[0010] FIG. 2 is a perspective view illustrating a state where the vehicle door handle device according to the embodiment of this disclosure is attached to the vehicle door;

[0011] FIG. 3A is a perspective view of the vehicle door handle device according to the embodiment of this disclosure;

[0012] FIG. 3B is a perspective view of the vehicle door handle device according to the embodiment of this disclosure;

[0013] FIG. 4A is an exploded view of the vehicle door handle device according to the embodiment of this disclosure;

[0014] FIG. 4B is an exploded view of the vehicle door handle device according to the embodiment of this disclosure;

[0015] FIG. 5A is an exploded view of the vehicle door handle device according to the embodiment of this disclosure;

[0016] FIG. 5B is an exploded view of the vehicle door handle device according to the embodiment of this disclosure;

[0017] FIG. 6 is a sectional view of the vehicle door handle device according to the embodiment of this disclosure;

[0018] FIG. 7 is a sectional view of the vehicle door handle device according to the embodiment of this disclosure; and

[0019] FIG. 8 is a sectional view of the vehicle door handle device according to the embodiment of this disclosure.

DETAILED DESCRIPTION

[0020] The following describes an embodiment of this disclosure with reference to the drawings. In each drawing, a front side of a vehicle to which a door handle device is applied is indicated by the arrow Fr, a rear side of the vehicle is indicated by the arrow Rr, an outer side (hereinafter, referred to as “vehicle outer side” in some cases) in a vehicle width direction is indicated by the arrow Out, an inner side (hereinafter, referred to as “vehicle inner side” in some cases) in the vehicle width direction is indicated by the arrow In, an upper side of the vehicle is indicated by the arrow Up, and a lower side of the vehicle is indicated by the arrow Dw. In the following description, directions and mutual positional relations for the door handle device and constituents of the door handle device are based on the respective directions of the vehicle.

Configuration Example of Vehicle Door

[0021] FIG. 1 is a schematic diagram illustrating a configuration example of a vehicle door 50 to which the door handle device 10 has been applied, and illustrates the vehicle door 50 viewed from a vehicle outer side. FIG. 2 is a perspective view illustrating a state in which the door handle device 10 is attached to the vehicle door 50. In this embodiment, the door handle device is applied to driver's door. The vehicle door 50 is a swing-type door that opens and closes by rotating relative to

a vehicle body around a front end of the vehicle door **50**. The vehicle door **50** includes a door body portion **51** constituting a lower half of the vehicle door **50**, and a door sash **52** provided at an upper half of the vehicle door **50**. The door body portion **51** includes an outer panel **53** constituting an outer side surface of the door body portion **51**, an unillustrated inner panel fixed on an vehicle inner side of the outer panel **53**, and an unillustrated resin trim fixed to an inner side surface of the inner panel and constituting an inner side surface of the door body portion **51**. A panel opening **531** is provided at a position near a rear end portion of the outer panel **53** and near an upper end portion of the outer panel **53**. The panel opening **531** is a through hole that has a shape elongated in a front-rear direction and that penetrates in a vehicle width direction.

[0022] A door lock device **54** as well as the door handle device **10** are arranged in the door body portion **51**. The door lock device **54** is arranged on a lower side of the door handle device **10**. The door lock device **54** includes a latch mechanism. The latch mechanism of the door lock device **54** is configured to be switchable between a latching state of not allowing the vehicle door **50** to be opened by each of driving force of a driving force source and manual operation of a user and an unlatching state of allowing the vehicle door **50** to be opened by each of the driving force and the manual operation. Further, the door lock device **54** is configured to be switchable between an unlocking state of allowing the latch mechanism to be switched from the latching state to the unlatching state and a locking state of not allowing the latch mechanism to be switched from the latching state to the unlatching state.

[0023] The door lock device **54** may include “a member configured to switch the latch mechanism from the latching state to the unlatching state by a downward movement of at least a part of the member”. Examples of a door lock device used as such a door lock device **54** includes one disclosed in JP2023-117102A or JP2022-079282A. The door lock devices disclosed in these publications each include an open lever that is supported by a casing of the door lock device rotatably relative to the casing. These door lock devices are each configured in such a way that rotation of the open lever in a manner of causing one end of the open lever to move downward switches the latch mechanism from a latching state to an unlatching state. Also in the following description, “a member that switches the latch mechanism from the latching state to the unlatching state by a downward movement of at least a part of the member” is referred to as “open lever”.

[0024] In addition to this, an electronic control unit (ECU) that controls the door lock device **54** is mounted on the vehicle.

[0025] As illustrated in FIG. **1** and FIG. **2**, the door handle device **10** is attached at a position (near the rear end portion of the outer panel **53** and near the upper end portion of the outer panel **53**) that is on a vehicle inner side of the outer panel **53** and that overlaps with the panel opening **531** when viewed in the vehicle width direction.

Configuration of Door Handle Device

[0026] FIG. **3A** and FIG. **3B** are perspective views illustrating a configuration of the door handle device **10**. FIG. **4A** and FIG. **4B** are exploded views illustrating a configuration of a case assembly **11** of the door handle device **10**. FIG. **5A** and FIG. **5B** are exploded views illustrating a configuration of a cover assembly **12**. The door handle device **10** includes the case assembly **11** attached to a vehicle inner side of the outer panel **53**, and the cover assembly **12** attached to the case assembly **11** from a vehicle outer side of the outer panel **53**.

[0027] The case assembly **11** includes a case body **20**, an inner cover **21**, an outer peripheral pad **22**, a switch assembly **23**, a flap **24**, a bell crank **25**, a latch release lever **26**, and a key cylinder **27**.

[0028] The case body **20** is a member that forms the casing of the door handle device **10**. The case body **20** includes a frame portion **201**, an accommodation portion **202**, and a key cylinder attachment portion **203**. The case body **20** is made of a resin material, and the above-described respective portions of the case body **20** are formed integrally with each other.

[0029] The frame portion **201** is a portion that is arranged in such a way as to overlap with the panel opening **531** when viewed in the vehicle width direction. When viewed in the vehicle width

direction, the frame portion **201** has a shape elongated in the front-rear direction, and more specifically has the substantially same shape as the panel opening **531**.

[0030] The accommodation portion **202** is a portion configured to be able to accommodate the flap **24**, the switch assembly **23**, and the guide member **31** of the cover assembly **12** which are described below. The accommodation portion **202** has a structure of a rectangular-tube shape (which can also be said as a box shape whose both vehicle-width-direction ends are open) penetrating in the vehicle width direction. An upper portion **204** (a portion corresponding to an upper plate, a front plate, and a rear plate that constitute the rectangular tube or the box) of the accommodation portion **202** is provided in such a way as to extend obliquely upward to a vehicle inner side from an upper end portion (which can also be said as an upper edge) of the frame portion **201** or from the vicinity of this upper end portion. A lower portion **205** (a portion corresponding to a lower plate that constitutes the rectangular tube or the box) of the accommodation portion **202** is provided in such a way as to extend obliquely upward to a vehicle inner side from a lower end portion (which can also be said as a lower edge) of the frame portion **201** or from the vicinity of this lower end portion.

[0031] The upper portion **204** of the accommodation portion **202** is configured to rotatably support the below-described flap **24** via a flap support shaft **245**. Specifically, a rear end portion of the upper portion **204** of the accommodation portion **202** includes a rear-side shaft hole **206** as a through hole that is substantially parallel to the front-rear direction and through which the flap support shaft **245** can be inserted from a rear side. A first support shaft attachment portion **207** is provided on a vehicle-inner-side surface of the upper portion **204** of the accommodation portion **202**, and has a projection shape protruding to a vehicle inner side. The first support shaft attachment portion **207** includes a front-side shaft hole **208** as a hole coaxial with the rear-side shaft hole **206** and through which (or into which) a distal end portion (front end portion) of the flap support shaft **245** can be inserted.

[0032] In addition to this, the upper portion **204** of the accommodation portion **202** includes a first stopper **209** defining a movable range of the flap **24**. The first stopper **209** is a portion having a projection-shaped structure that protrudes to a vehicle inner side from the vehicle-inner-side surface of the upper portion **204** of the accommodation portion **202**.

[0033] The key cylinder attachment portion **203** is a portion configured in such a way that the key cylinder **27** can be attached to the key cylinder attachment portion **203**. The key cylinder attachment portion **203** is provided at or near the rear end portion of the frame portion **201** in such a way as to be on a rear side of the accommodation portion **202**. The key cylinder attachment portion **203** has a cylindrical structure extending obliquely downward to a vehicle inner side from the frame portion **201**.

[0034] The inner cover **21** is a member attached on a vehicle inner side of the accommodation portion **202** of the case body **20**. The inner cover **21** has a substantially plate-shaped structure. The inner cover **21** attached on a vehicle inner side of the accommodation portion **202** closes a vehicle-inner-side opening of the accommodation portion **202**.

[0035] The outer peripheral pad **22** is a member attached on a vehicle outer side of the frame portion **201**. The outer peripheral pad **22** is a frame-shaped portion including an opening (a through hole penetrating in the vehicle width direction) that is the substantially same shape and size as those of the panel opening **531**. The outer peripheral pad **22** is arranged on a vehicle inner side of the outer panel **53** in such a way as to be along an inner peripheral edge portion of the panel opening **531**.

[0036] The switch assembly **23** is an assembly that includes a base member **231**, a switch **232**, a light source **233**, and a light guide member **234**. The base member **231** is a member to which the switch **232**, the light source **233**, and the light guide member **234** are attached. The switch **232** is a normally open type push button switch that is switched from an off state to an on state when a push button of the push button switch is pushed. The light source **233** and the light guide member **234**

are configured in such a way as to be able to emit light into an internal space of a fingertip guide portion **312** of the below-described guide member **31**. The switch assembly **23** is fixed to the vehicle-inner-side surface of the upper portion **204** of the accommodation portion **202** of the case body **20**. The switch assembly **23** is arranged in such a way that the push button of the switch **232** faces to a vehicle inner side (in such a way as to be switched from the off state to the on state when the push button is pushed from an vehicle inner side). The switch assembly **23** is arranged on a front side of the flap **24**. The switch **232** and the light source **233** of the switch assembly **23** are connected to the ECU.

[0037] The flap **24** is a member that is manually operated by a user when the user switches the latch mechanism of the door lock device **54** at the normal time from the latching state to the unlatching state. The flap **24** includes a first finger hook portion **241**, a push portion **242**, a second support shaft attachment portion **243**, and a second stopper **244**. The flap **24** is a member formed by injection molding of a resin material, and the respective portions of the flap **24** are formed integrally with each other.

[0038] The first finger hook portion **241** is a portion having a box shape (which can also be said as a shell shape open on a vehicle inner inside and on a lower side) that is open on a vehicle inner inside and a lower side and that is closed on a vehicle outer side, an upper side, a front side, and a rear side. The push portion **242** is a portion for switching a state of the switch **232** of the switch assembly **23** (described below). The push portion **242** has a rod-shaped structure that protrudes to a substantially front side from a position (or from the vicinity of this position) that is at a vehicle-inner-side end of a front end surface of the first finger hook portion **241** and that is at a lower end of the front end surface of the first finger hook portion **241**. The second support shaft attachment portion **243** is provided on an upper side of the first finger hook portion **241**, and is configured in such a way that the flap support shaft **245** can be attached to the second support shaft attachment portion **243**. Specifically, the second support shaft attachment portion **243** has a projection-shaped structure that protrudes to an upper side from an upper surface of the first finger hook portion **241**. The second support shaft attachment portion **243** includes an shaft hole that penetrates in the substantially front-rear direction and through which the flap support shaft **245** can be inserted.

[0039] The flap **24** is arranged at a position that is inside the accommodation portion **202** of the case body **20** and that is on a vehicle inner side of the upper portion **204** of the accommodation portion **202**. The flap **24** is supported by the case body **20** via the flap support shaft **245** rotatably relative to the case body **20**. Specifically, the flap support shaft **245** is inserted from a rear side through the shaft hole of the second support shaft attachment portion **243** of the flap **24** and the rear-side shaft hole **206** and the front-side shaft hole **208** of the accommodation portion **202**. A metal round bar is used as the flap support shaft **245**. Since the center line (axis) of the flap support shaft **245** is substantially parallel to the front-rear direction, the flap **24** can rotate relative to the case body **20** around the axis substantially parallel to the front-rear direction. Since the first finger hook portion **241** of the flap **24** is located on a lower side of the flap support shaft **245**, the first finger hook portion **241** can reciprocate in the substantially vehicle width direction, in a pendulum manner of swinging around the flap support shaft **245**.

[0040] The flap **24** is movable between an initial position and an actuation position by rotating relative to the case body **20**. The initial position is a position when the first finger hook portion **241** is located at a vehicle-inner-side end of the movable range. The initial position can also be said as a position when the first finger hook portion **241** is located at the position farthest from the outer panel **53**. Specifically, the initial position is a position at which the second stopper **244** of the flap **24** contacts with a vehicle-outer-side surface of the inner cover **21**. The actuation position is a position when the first finger hook portion **241** is located at a vehicle-outer-side end of the movable range. Specifically, the actuation position is a position at which a vehicle-outer-side surface of the first finger hook portion **241** contacts with the first stopper **209** of the accommodation portion **202**. The flap **24** located at the initial position does not push the push button of the switch **232** of the

switch assembly **23** (i.e., keeps the switch **232** in the off state), and the flap **24** moved from the initial position to the actuation position pushes the push button of the switch **232** (i.e., switches the switch **232** from the off state to the on state).

[0041] The flap **24** is constantly elastically biased toward the initial position by a flap biasing member **246**. Thus, the flap **24** is kept at the initial position by the biasing force of the flap biasing member **246** in a state where no external force other than that from the flap biasing member **246** is applied to the flap **24**. A torsion coil spring whose both ends each include an arm is used as the flap biasing member **246**. The flap biasing member **246** is arranged in such a way that the flap support shaft **245** is inserted through a coil-shaped portion of the flap biasing member **246**, the one arm contacts with the case body **20**, and the other arm contacts with the flap **24**.

[0042] The bell crank **25** is a member that is linked to the door lock device **54** via a rod member **28** (refer to FIG. **8**). The bell crank **25** is configured in such a way as to move in conjunction with the latch release lever **26** and thereby switch the latch mechanism of the door lock device **54** from the latching state to the unlatching state via the rod member **28**. The bell crank **25** includes a substantially rod-shaped rod coupling portion **251** that extends in a direction substantially parallel to the vehicle width direction, a projection-shaped latch release lever engagement portion **252** that protrudes to a lower side from a vehicle-outer-side end portion of the rod coupling portion **251**, and a shaft portion **253** that is provided at the vehicle-outer-side end portion of the rod coupling portion **251** and that protrudes in the front-rear direction. Thus, the bell crank **25** has a substantially “L” shape when viewed in the front-rear direction.

[0043] The bell crank **25** is arranged on a rear side of the accommodation portion **202** of the case body **20**. The bell crank **25** is supported at the vehicle-outer-side end portion of the rod coupling portion **251** via the shaft portion **253** rotatably relative to the case body **20** around the axis that is substantially parallel to the front-rear direction. Thus, the bell crank **25** can rotate relative to the case body **20** in such a way that the rod coupling portion **251** moves in a substantially up-down direction and the bell crank engagement portion **261** moves in the substantially vehicle width direction. The bell crank **25** can move between an initial position and an actuation position by rotating relative to the case body **20**. The initial position of the bell crank **25** is a position at which a distal end portion (vehicle-inner-side end portion) of the rod coupling portion **251** is located at an upper end of a movable range of this distal end portion and a distal end portion (lower end portion) of the latch release lever engagement portion **252** is located at a vehicle-inner-side end of the movable range of this distal end portion. The actuation position of the bell crank **25** is a position where the distal end portion of the rod coupling portion **251** is located at a lower end of the movable range of this distal end portion and the distal end portion of the latch release lever engagement portion **252** is located at a vehicle-outer-side end of the movable range of this distal end portion.

[0044] The bell crank **25** is constantly elastically biased toward the initial position by a bell crank biasing member **254**. For this reason, the bell crank **25** is kept at the initial position in a state where no external force other than that of the bell crank biasing member **254** is applied to the bell crank **25**. A torsion coil spring whose both ends each include an arm is used as the bell crank biasing member **254**. The bell crank biasing member **254** is arranged in such a way that the shaft portion **253** of bell crank **25** is inserted through a coil-shaped portion of the bell crank biasing member **254**, the one arm engages with the bell crank **25**, and the other arm engages with case body **20**.

[0045] The latch release lever **26** is a member that is manually operated by a user when the user switches the latch mechanism of the door lock device **54** from the latching state to the unlatching state in an emergency or the like. Examples of the emergency include a case where the latch mechanism of the door lock device **54** cannot be switched from the latching state to the unlatching state by operating the flap **24**, or a case where the flap **24** cannot be manually operated. More specific examples of the emergency include a case where the ECU does not work because a charged amount of the battery is small.

[0046] The latch release lever **26** is a curved rod-shaped member having a substantially “U” shape (or a substantially “C” shape) when viewed in the up-down direction. The latch release lever **26** includes the bell crank engagement portion **261**, a second finger hook portion **262**, and a third stopper **263**. The bell crank engagement portion **261** is provided at an intermediate position in the latch release lever **26** in the front-rear direction, and is a portion having a projection-shaped structure that protrudes to a vehicle outer side. The second finger hook portion **262** is provided at or near a front end portion of the latch release lever **26**, and is a portion configured in such a way that a user can hook his or her finger on the second finger hook portion **262**. The second finger hook portion **262** has a plate shape. The third stopper **263** is a portion that defines the below-described initial position of the latch release lever **26**. The third stopper **263** is a portion that is provided at a position on a vehicle inner side of the second finger hook portion **262** and that has a surface facing to a vehicle inner side.

[0047] The latch release lever **26** is arranged at a position that is on a rear side of the accommodation portion **202** of the case body **20** and that overlaps with the bell crank **25** when viewed in the up-down direction, while the latch release lever **26** is oriented in such a way that an opening of the “U” shape (or “C” shape) is located on a vehicle outer side. The latch release lever **26** is supported by the case body **20** rotatably relative to the case body **20** around an axis that is substantially parallel to the up-down direction and that is at an end portion on a rear side and a vehicle outer side in the latch release lever **26**. Thus, the second finger hook portion **262** and the bell crank engagement portion **261** of the latch release lever **26** can reciprocate in the substantially vehicle width direction. The latch release lever **26** can move between the initial position and an actuation position by rotating relative to the case body **20**. The initial position of the latch release lever **26** is a position when the second finger hook portion **262** is located at a vehicle-inner side end of a movable range of the second finger hook portion **262**. Specifically, the initial position of the latch release lever **26** is a position when the third stopper **263** of the latch release lever **26** contacts with the case body **20**. The actuation position of the latch release lever **26** is a position when the second finger hook portion **262** is located on a vehicle outer side of the initial position.

[0048] The bell crank **25** and the latch release lever **26** are arranged at positions where the bell crank **25** and the latch release lever **26** overlap with each other when viewed in the up-down direction. The bell crank **25** and the latch release lever **26** are arranged in such a way that the bell crank engagement portion **261** of the latch release lever **26** is located on a vehicle inner side of the latch release lever engagement portion **252** of the bell crank **25**, and the latch release lever engagement portion **252** of the bell crank **25** and the bell crank engagement portion **261** of the latch release lever **26** overlap with each other when viewed in the vehicle width direction.

[0049] When the bell crank **25** and the latch release lever **26** are arranged as described above, the latch release lever engagement portion **252** of the bell crank **25** elastically biases the bell crank engagement portion **261** of the latch release lever **26** to an vehicle inner side by the biasing force of the bell crank biasing member **254**. Thus, the latch release lever **26** is constantly elastically biased to the initial position by the biasing force of the bell crank biasing member **254** transmitted via the bell crank engagement portion **261**. Accordingly, the latch release lever **26** is kept at the initial position when no biasing force other than the biasing force of the bell crank biasing member **254** is applied to the latch release lever **26**. When the latch release lever **26** moves from the initial position toward the actuation position, the latch release lever engagement portion **252** of the bell crank **25** is pushed to a vehicle outer side by the bell crank engagement portion **261** of the latch release lever **26**, and thus moves from the initial position to the actuation position against the biasing force of the bell crank biasing member **254**. In this manner, the bell crank **25** and the latch release lever **26** are mechanically linked to each other in such a way that when the latch release lever **26** is located at the initial position, the bell crank **25** is also located at the initial position, and when the latch release lever **26** is located at the actuation position, the bell crank **25** is also located at the actuation position.

[0050] The key cylinder **27** is a member that can be manually operated by a user using an adapted key. The key cylinder **27** is mechanically linked to the door lock device **54** in such a way as to be switchable between a locking state and an unlocking state. The locking state is a state of not allowing the latch mechanism of the door lock device **54** to be switched from the latching state to the unlatching state, and the unlocking state is a state of allowing the latch mechanism to be switched from the latching state to the unlatched state. A configuration of the key cylinder **27** and a configuration of the link between the key cylinder **27** and the door lock device **54** are not particularly limited, and any conventionally known configurations can be applied to these configurations. The key cylinder **27** is supported by the case body **20** by being inserted into an inside of the cylindrical shape of the key cylinder attachment portion **203**.

[0051] The rod member **28** is a substantially rod-shaped member. The rod member **28** is arranged in such a way that an elongated direction of the rod member **28** is oriented in the substantially up-down direction. An upper end portion of the rod member **28** is coupled to a vehicle-inner-side end portion of the rod coupling portion **251** of the bell crank **25**. A lower end portion of the rod member **28** is coupled to the open lever of the door lock device **54**, or can freely engage with and freely disengage from the open lever. When the bell crank **25** moves from the initial position to the actuation position, the rod member **28** is pushed by the rod coupling portion **251** of the bell crank **25** and thus moves to a lower side, and thereby pushes down the open lever of the door lock device **54**. Thereby, the latch mechanism of the door lock device **54** is switched from the latching state to the unlatching state. In this manner, the rod member **28** is mechanically linked to the bell crank **25** and the door lock device **54** in such a way as to switch the latch mechanism of the door lock device **54** from the latching state to the unlatching state when the bell crank **25** moves from the initial position to the actuation position.

[0052] The cover assembly **12** includes an exterior member **30** and a guide member **31**.

[0053] The exterior member **30** is a member located on the most vehicle outer side in the door handle device **10**, and forms a part of an exterior of the door handle device **10**. The exterior member **30** is a plate-shaped member having the substantially same shape as that of the panel opening **531** when viewed in the vehicle width direction, and is configured to be allowed to be fitted into the panel opening **531** of the vehicle door **50**. The exterior member **30** includes a fingertip insertion hole **301** into which a user can insert his or her fingertip from a vehicle outer side. The fingertip insertion hole **301** is a through hole that has a shape elongated in the front-rear direction and that penetrates in the vehicle width direction. In a state where the exterior member **30** is attached to the outer panel **53** of the vehicle door **50**, the exterior member **30** is exposed to a vehicle outer side from the panel opening **531** in such a way that a vehicle-outer-side surface of the exterior member **30** is substantially flush with a vehicle-outer-side surface of the outer panel **53** of the vehicle door **50**.

[0054] The guide member **31** is a member that forms the casing of the door handle device **10**. The guide member **31** is a member to which the exterior member **30** is attached. The guide member **31** includes a body portion **311**, a fingertip guide portion **312**, and a key cover attachment portion **313**. The body portion **311** is a portion that is arranged on a vehicle inner side of the exterior member **30** in such a way as to overlap with the exterior member **30**. The body portion **311** has an external plate-shape that is the substantially same shape and size as those of the exterior member **30**. Further, the body portion **311** includes an opening that has the substantially same shape and size as those of the fingertip insertion hole **301** of the exterior member **30**.

[0055] The fingertip guide portion **312** is a portion for guiding user's fingertip inserted from the panel opening **531** into an inside (more specifically, a vehicle-inner-side surface of the first finger hook portion **241** of the flap **24** arranged in the inside of the accommodation portion **202**) of the accommodation portion **202** of the case body **20**. The fingertip guide portion **312** is a portion provided in such a way as to extend obliquely upward to an vehicle inner side from the vicinity of a lower end portion (the vicinity of a lower edge) of the body portion **311**. The fingertip guide

portion **312** has a shallow box-shaped structure (which can also be said as a tray-shaped structure) that is open on a vehicle outer side, an upper side, and a rear side and is closed on a lower side and a front side. The fingertip guide portion **312** includes an inclined surface (guide surface) that faces to an upper side and a vehicle outer side and forms a slope extending upward to a vehicle inner side.

[0056] The key cover attachment portion **313** is a portion configured to allow the key cover **32** to be attached to and detached from the key cover attachment portion **313**, and is provided at a position that is on a rear side of the fingertip guide portion **312** and that is near the rear end portion of the guide member **31**. The key cover attachment portion **313** is, for example, a plate-shaped portion including a through hole that penetrates in the vehicle width direction (or can be said as a frame-shaped portion that includes the opening on an inner peripheral side).

[0057] The key cover **32** is a member that is arranged on a vehicle outer side of an end surface (a surface on which a key hole is formed) of the key cylinder **27** attached to the key cylinder attachment portion **203** of the case body **20** and that thereby covers the end surface of the key cylinder **27** (in other words, hides the end surface of the key cylinder **27**). The key cover **32** is a member that can be attached to and detached from the key cover attachment portion **313**. For example, the key cover **32** has a flat-plate shape, and is configured to be attachable to and detachable from the key cover attachment portion **313** by a snap-fit structure.

[0058] FIG. **6** a sectional view taken along a plane substantially perpendicular to the up-down direction, and illustrates a state where the door handle device **10** is attached to the outer panel **53**. As illustrated in FIG. **6**, in a state where the cover assembly **12** is attached to the case assembly **11**, the case body **20** of the case assembly **11** and the guide member **31** of the cover assembly **12** form the casing of the door handle device **10**. The casing of the door handle device **10** includes a concave portion that is formed by the fingertip guide portion **312** of the guide member **31** and the second finger hook portion **262** of the latch release lever **26**, that is open to a vehicle outer side, and into which a user can insert his or her fingertip from a vehicle outer side.

[0059] The key cover **32** attached to the key cover attachment portion **313** covers a vehicle-outer-side end surface of the key cylinder **27**, and is exposed to a vehicle outer side through the fingertip insertion hole **301** of the exterior member **30**. The second finger hook portion **262** of the latch release lever **26** attached to the case assembly **11** is located at a position that is on a front side of the key cylinder attachment portion **203** and that is on a vehicle inner side of the key cylinder attachment portion **203**. The second finger hook portion **262** of the latch release lever **26** attached to the case assembly **11** is exposed to a vehicle-outer side and to an obliquely front side through the fingertip insertion hole **301** of the exterior member **30**. In this manner, the key cover **32** and the latch release lever **26** form a part of the exterior of the door handle device **10**. Particularly, the second finger hook portion **262** of the latch release lever **26** is located at a rear end portion (i.e., the concave portion into which a user can insert his or her fingertip) of the fingertip guide portion **312** of the guide member **31**, and closes an opening at the rear end portion of the fingertip guide portion **312**. The latch release lever **26** is located at a rear end portion of the rear end portion (i.e., the concave portion into which a user can insert his or her fingertip) of the fingertip guide portion **312**, and forms a rear surface portion (a surface located at the rear end portion of the concave portion and facing to a substantially front side) included in an inner peripheral surface of the concave portion and facing to a substantially front side. More specifically, a portion included in the rear surface portion and on a vehicle outer side is formed by the second finger hook portion **262**. A portion (the portion on a vehicle inner side of the second finger hook portion **262**) included in the rear surface portion and on a vehicle inner side is formed by a portion included in the latch release lever **26** and on a vehicle inner side of the second finger hook portion **262**. The rear surface portion is an example of an inner peripheral surface formation portion in this disclosure.

[0060] When viewed in the up-down direction, a vehicle-outer-side surface of the key cover **32** and a front-side surface of the second finger hook portion **262** of the latch release lever **26** are inclined

from the front-rear direction in such a way as to be more shifted to a vehicle inner side on a more front side. However, an inclination angle of the vehicle-outer-side surface of the key cover **32** made with a straight line parallel to the front-rear direction is different from that of the front-side surface of the latch release lever **26**. Specifically, the angle made between the straight line parallel to the front-rear direction and the vehicle-outer-side surface of the key cover **32** is smaller than the angle made between the straight line parallel to the front-rear direction and the front-side surface of the latch release lever **26**. The vicinity of a vehicle-outer-side end portion of the front-side surface of the latch release lever **26** is curved, and is connected in series to the vehicle-outer-side surface of the key cover **32** (in other words, is substantially flush with the vehicle-outer-side surface of the key cover **32**). According to such a configuration, the second finger hook portion **262** of the latch release lever **26** and the key cover **32** appear to be mutually integrated (continuous). Thereby, the aesthetic appearance can be improved (the design quality can be improved).

[0061] The front-side surface of the second finger hook portion **262** of the latch release lever **26** can be said to be inclined from the vehicle width direction in such a way as to protrude to a front side on a more vehicle inner side. A vehicle-inner-side end portion of the second finger hook portion **262** of the latch release lever **26** protrudes to a front side more than a portion included in the latch release lever **26** and located on a vehicle inner side of the second finger hook portion **262**. Thus, a user can hook his or her fingertip on the vehicle-inner-side end portion of the second finger hook **262** of the latch release lever **26**. The vehicle-inner-side end portion of the second finger hook portion **262** and the portion located on an vehicle inner side of this vehicle-inner-side end portion in the latch release lever **26** are discontinuous with each other. However, the vehicle-inner-side end portion of the second finger hook portion **262** of the latch release lever **26** exists at a position deep from the fingertip insertion hole **301** of the guide member **31**, and is thus not noticeable.

[0062] According to this embodiment, the latch release lever **26** is located at the rear end portion of the concave portion (the fingertip guide portion **312**) into which a fingertip can be inserted, and forms the rear surface portion (the surface facing to a front side) in the inner peripheral surface of this concave portion so that the latch release lever **26** is less noticeable. For this reason, a user is prevented or suppressed from mistaking the latch release lever **26** for “a member normally used for unlatching operation of the vehicle door **50**”. According to such a configuration, a boundary between the latch release lever **26** and other portions is less noticeable (i.e., an edge of the design portion can be less noticeable) so that the aesthetic appearance (the design quality) can be improved.

[0063] The vehicle-outer-side surface of the key cover **32** attached to the key cover attachment portion **313** is located at a position shifted from the vehicle-outer-side surface of the exterior member **30** to a vehicle inner side (in other words, at a position deeper to a vehicle inner side than the vehicle-outer-side surface of the exterior member **30**). In other words, the vehicle-outer-side surface of the exterior member **30** is discontinuous with the vehicle-outer-side surface of the key cover **32** attached to the key cover attachment portion **313**. With such a configuration, the vehicle-outer-side end portion of the latch release lever **26** is also located at a position shifted from the vehicle-outer-side surface of the exterior member **30** to a vehicle inner side (in other words, at a position deeper to a vehicle inner side than the vehicle-outer-side surface of the exterior member **30**). Thus, an advantageous effect of making the latch release lever **26** less noticeable can be further enhanced. Further, according to such a configuration, the boundary between the latch release lever **26** and other portions is located at a position shifted from the vehicle-inner-side surface of the exterior member **30** to a vehicle inner side. For this reason, an advantageous effect of making this boundary less noticeable can be enhanced.

[0064] In this embodiment, the bell crank **25** and the latch release lever **26** are arranged at positions that are on an upper side of the key cylinder **27** and that overlap with the key cylinder **27** when viewed in the up-down direction. Such a configuration allows a dimension of the door handle device **10** in the front-rear direction to be reduced as compared with a configuration in which the

bell crank **25** and the latch release lever **26** are arranged at positions shifted from each other in the front-rear direction in such a way as not to overlap with the key cylinder **27** when viewed in the up-down direction. The key cylinder **27** is arranged to be inclined in such a way that the vehicle-inner-side end portion of the key cylinder **27** is located on a lower side of the vehicle-outer-side end portion of the key cylinder **27**. Thus, according to such a configuration, the bell crank **25**, the latch release lever **26**, and the key cylinder **27** can be arranged in such a way that the bell crank **25** and the latch release lever **26** overlap with the key cylinder **27** in the up-down direction without interfering with the key cylinder **27**.

Actuation of Door Handle Device

[0065] Next, actuation of the door handle device **10** is described. When the flap **24** is located at the initial position, the push portion **242** is separated from the switch **232** of the switch assembly **23** to a vehicle inner side, and thus, the push button of the switch **232** is not pushed. For this reason, when the flap **24** is located at the initial position, the switch **232** is kept in the off state. Meanwhile, when the flap **24** moves from the initial position toward the actuation position, the push portion **242** pushes the push button of the switch **232**. Then, when the flap **24** is located at the actuation position, the push button of the switch **232** is in a state of being pushed by the push portion **242**, and the switch **232** is in the on state.

[0066] A user inserts his or her fingertip into the panel opening **531** while making his or her palm face to an upper side, and hooks the fingertip on the lower end portion of the first finger hook portion **241** of the flap **24** from a lower side. Then, in this state, the user pulls the first finger hook portion **241** to a vehicle outer side so that the first finger hook portion **241** moves to a vehicle outer side against the biasing force of the flap biasing member **246**. Thus, the push portion **242** moves to a vehicle outer side together with the first finger hook portion **241**, and thereby pushes the push button of the switch **232** assembled in the switch assembly **23**. As a result, the switch **232** is switched from the off state to the on state.

[0067] When the ECU detects that the switch **232** is switched from the off state to the on state, the ECU switches the latch mechanism of the door lock device **54** from the latching state to the unlatching state. When the door lock device **54** is in the locking state, the ECU switches the door lock device **54** to the unlocking state, and then switches the door lock device **54** to the unlatching state. The user successively pulls the first finger hook portion **241** to a vehicle outer side, and can thereby open the vehicle door **50**. Then, when the user releases the fingertip from the first finger hook portion **241**, the flap **24** moves to the initial position by the biasing force of the flap biasing member **246** so that the switch **232** is switched from the on state to the off state. In this manner, at the normal time, the user can open the vehicle door **50** (switch the door lock device **54** from the latching state to the unlatching state) by operating the flap **24**.

[0068] In an emergency such as a case where the ECU is not working, a user can manually operate the latch release lever **26** to thereby switch the latch mechanism of the door lock device **54** from the latching state to the unlatching state. FIG. 7 and FIG. 8 are sectional views illustrating actuation of the latch release lever **26** and the bell crank **25**.

[0069] When the door lock device **54** is in the locking state, a user detaches the key cover **32**, and switches the door lock device **54** from the locking state to the unlocking state by using the key and thus operating the key cylinder **27**. Then, the user moves the latch release lever **26** from the initial position to the actuation position as illustrated in FIG. 7. In FIG. 7, the latch release lever **26** located at the initial position is depicted by the solid lines, and the latch release lever **26** located at the actuation position is depicted by the two-dot chain lines. At this time, the user can pull the second finger hook portion **262** out to a vehicle outer side by hooking his or her fingertip on the vehicle-inner-side end portion of the second finger hook portion **262** of the latch release lever **26**.

[0070] The movement of the latch release lever **26** from the initial position to the actuation position causes the bell crank engagement portion **261** of the latch release lever **26** to push the latch release lever engagement portion **252** of the bell crank **25** to a vehicle outer side. Thereby, the bell crank

25 rotates around the shaft portion **253** and thus move from the initial position to the actuation position. as illustrated in FIG. **8**, At this time, the rod coupling portion **251** of the bell crank **25** pushes the rod member **28** to a lower side. In FIG. **8**, the bell crank **25** located at the initial position is depicted by the solid lines, and the bell crank **25** located at the actuation position is depicted by the two-dot chain line. Then, pushing the open lever of the door lock device **54** to a lower side by the rod member **28** causes the latch mechanism of the door lock device **54** to be switched from the latching state to the unlatching state. In this manner, the latch release lever **26** is mechanically linked to the door lock device **54** via the bell crank **25** and the rod member **28** in such a way as to switch the latch mechanism of the door lock device **54** from the latching state to the unlatching state by being moved from the initial position to the actuation position.

[0071] In this manner, the door handle device **10** is configured to allow the latch mechanism of the door lock device **54** to be switched from the latching state to the unlatching state via the rod member **28**. Since the rod member **28** is less expensive than a wire, component cost of the door handle device **10** can be reduced as compared with a configuration in which the door lock device **54** is linked via the wire.

Outline of Embodiment

[0072] (1) A vehicle door handle device **10** attached to a vehicle door **50** and including a door lock device **54** that is switchable between a latching state of not allowing the vehicle door **50** to be opened and an unlatching state of allowing the vehicle door **50** to be opened, the vehicle door handle device **10** including: [0073] a casing (a case body **20** and a guide member **31**) open on a vehicle outer side and including a concave portion (a fingertip guide portion **312**) into which a user can insert his or her finger from a vehicle outer side; and [0074] a latch release lever **26** attached to the casing (the case body **20** and the guide member **31**) in such a way as to be movable between an initial position and an actuation position, the latch release lever **26** being linked to the door lock device **54** in such a way as to switch the door lock device **54** from the latching state to the unlatching state by moving from the initial position to the actuation position, wherein [0075] an inner peripheral surface of the concave portion (the fingertip guide portion **312**) includes a rear surface portion located on a vehicle inner side of a vehicle-outer-side surface of the vehicle door **50** and located at a rear end portion of the concave portion (the fingertip guide portion **312**), the rear surface portion facing to a substantially front side, and [0076] the latch release lever **26** forms at least a part of the rear surface portion.

[0077] According to this embodiment, the latch release lever **26** is located at the rear end portion of the concave portion (the fingertip guide portion **312**) of the casing (the case body **20** and the guide member **31**), and forms the rear surface portion that is a surface included in the inner peripheral surface of the concave portion (the fingertip guide portion **312**) and facing to a front side. For this reason, the latch release lever **26** is less noticeable. Thus, with such a configuration, a user can be prevented or suppressed from mistakenly recognizing that “the latch release lever **26** is a member used for opening the vehicle door **50** at the normal time (the latch release lever **26** is a member that is normally used for opening the vehicle door **50**)”. With such a configuration, the boundary between the latch release lever **26** and other portions is less noticeable (i.e., the edge of the design portion can be made less noticeable) so that the aesthetic appearance (design quality) can be improved.

[0078] (2) The vehicle door handle device **10** according to this embodiment includes: [0079] a key cylinder **27** linked to the door lock device **54** in such a way as to be able to switch the door lock device **54** between a locking state and an unlocking state. The locking state is a state of not allowing the door lock device **54** to be switched from the latching state to the unlatching state. The unlocking state is a state of allowing the door lock device **54** to be switched from the latching state to the unlatching state, wherein [0080] the latch release lever **26** and the key cylinder **27** overlap with each other when viewed in an up-down direction.

[0081] According to such a configuration, a dimension of the door handle device **10** in a front-rear

direction can be prevented or suppressed from being increased.

[0082] (3) The vehicle door handle device **10** according to the present embodiment includes:

[0083] a bell crank **25** that is arranged to be movable between an initial position and an actuation position relative to the casing (the case body **20** and the guide member **31**), and that is linked to the latch release lever **26** in such a way that when the latch release lever **26** is located at the initial position, the bell crank **25** is located at the initial position, and when the latch release lever **26** is located at the actuation position, the bell crank **25** is located at the actuation position; and [0084] a rod member **28** that is linked to the bell crank **25** and the door lock device **54** in such a way as to switch the door lock device **54** from the latching state to the unlatching state when the bell crank **25** moves from the initial position to the actuation position.

[0085] Since the rod member **28** is a substantially rod-shaped member and is less expensive than a wire, the component cost of the door handle device **10** can be reduced as compared with a configuration in which the door lock device **54** is linked via the wire.

[0086] (4) In the vehicle door handle device **10** according to this embodiment. [0087] the bell crank **25** is configured in such a way as to move the rod member **28** to a lower side by moving from the initial position to the actuation position.

[0088] A conventionally known door lock device is configured in such a way as to include an open lever and be switched from a latching state to an unlatching state by movement of the open lever to a lower side. The vehicle door handle device **10** according to this embodiment can be applied to a vehicle door including such a door lock device.

[0089] Although the embodiment of this disclosure is described above, this disclosure is not limited to the above-described embodiment. This disclosure can be variously modified without departing from the essence of this disclosure, and these modifications are also encompassed in the technical scope of this disclosure.

[0090] For example, the rear surface portion of the concave portion in the above-described embodiment is formed by only the latch release lever **26**, but there is no limitation to such a configuration. For example, the fingertip guide portion **312** of the guide member **31** may include a portion that forms “the rear surface portion of the concave portion”, and the second finger hook portion **262** and the fingertip guide portion **312** may form “the rear surface portion of the concave portion”. In short, the latch release lever **26** may form at least a part of “the rear surface portion of the concave portion”.

[0091] The vehicle door handle device **10** in the above-described embodiment includes the key cylinder **27**, but the vehicle door handle device **10** does not need to include the key cylinder **27**. The switch assembly **23** in the above-described embodiment is arranged on a front side of the flap **24**, but a positional relation between the flap **24** and the switch assembly **23** is not limited to that in the above-described embodiment.

[0092] A vehicle door handle device according to this disclosure is attached to a vehicle door including a door lock device that is switchable between a latching state of not allowing the vehicle door to be opened and an unlatching state of allowing the vehicle door to be opened. The vehicle door handle device includes a casing and a latch release lever. The casing is open on a vehicle outer side, and includes a concave portion into which a user can insert his or her finger from a vehicle outer side. The latch release lever is attached to the casing in such a way as to be movable between an initial position and an actuation position. The latch release lever is linked to the door lock device in such a way as to switch the door lock device from the latching state to the unlatching state by moving from the initial position to the actuation position. An inner peripheral surface of the concave portion includes an inner peripheral surface formation portion that is located on a vehicle inner side of a vehicle-outer-side surface of the vehicle door and that is located at an end portion of the concave portion on one side in a front-rear direction of the vehicle door. The inner peripheral surface formation portion faces to a side substantially opposite to the one side in the front-rear direction of the vehicle door. The latch release lever forms at least a part of the inner peripheral

surface formation portion.

[0093] In the vehicle door handle device according to this disclosure, the latch release lever is located at the end portion on the one side in the front-rear direction of the vehicle door, and forms at least a part of the inner peripheral surface formation portion being a surface included in an inner peripheral surface of the concave portion and facing to the side substantially opposite to the one side. For this reason, the latch release lever is less noticeable. Thus, a user can be prevented or suppressed from mistakenly recognizing that “the latch release lever is a member to be used for opening the vehicle door at a normal time (the latch release lever is a member that is normally used for opening the vehicle door)”. With such a configuration, a boundary between the latch release lever and other portions is less noticeable (i.e., an edge of a design portion of the latch release lever can be made less noticeable), and thus the aesthetic appearance (design quality) can be improved.

[0094] In the vehicle door handle device described above, the inner peripheral surface formation portion may be located at a rear end portion of the concave portion, and may face to a substantially front side.

[0095] The vehicle door handle device described above may include a key cylinder. The key cylinder may be linked to the door lock device in such a way as to be able to switch the door lock device between a locking state and an unlocking state. The locking state is a state of not allowing the door lock device to be switched from the latching state to the unlatching state. The unlocking state is a state of allowing the door lock device to be switched from the latching state to the unlatching state. The latch release lever and the key cylinder may overlap with each other when viewed in an up-down direction.

[0096] With such a configuration, a dimension of the door handle device in a front-rear direction can be prevented or suppressed from being increased.

[0097] The vehicle door handle device described above may include a bell crank and a rod member. The bell crank may be arranged to be movable between an initial position and an actuation position relative to the casing. The bell crank may be linked to the latch release lever in such a way that, when the latch release lever is located at the initial position, the bell crank is located at the initial position, and when the latch release lever is located at the actuation position, the bell crank is located at the actuation position. The rod member may be linked to the bell crank and the door lock device in such a way as to switch the door lock device from the latching state to the unlatching state when the bell crank moves from the initial position to the actuation position.

[0098] Since the rod member is a substantially rod-shaped member and is less expensive than a wire, the component cost of the door handle device can be reduced as compared with a configuration in which the door lock device is linked via the wire.

[0099] In the vehicle door handle device described above, the bell crank may be configured in such a way as to move the rod member to a lower side by moving from the initial position to the actuation position.

[0100] A conventionally known door lock device is configured in such a way as to include an open lever and be switched from a latching state to an unlatching state by movement of the open lever to a lower side. The vehicle door handle device of the fifth aspect can be applied to a vehicle door including such a door lock device.

[0101] The principles, preferred embodiment and mode of operation of the present invention have been described in the foregoing specification. However, the invention which is intended to be protected is not to be construed as limited to the particular embodiments disclosed. Further, the embodiments described herein are to be regarded as illustrative rather than restrictive. Variations and changes may be made by others, and equivalents employed, without departing from the spirit of the present invention. Accordingly, it is expressly intended that all such variations, changes and equivalents which fall within the spirit and scope of the present invention as defined in the claims, be embraced thereby.

Claims

1. A vehicle door handle device to be attached to a vehicle door including a door lock device that is switchable between a latching state of not allowing the vehicle door to be opened and an unlatching state of allowing the vehicle door to be opened, the vehicle door handle device comprising: a casing being open on a vehicle outer side and including a concave portion into which a user can insert his or her finger from a vehicle outer side; and a latch release lever attached to the casing in such a way as to be movable between an initial position and an actuation position, the latch release lever being linked to the door lock device in such a way as to switch the door lock device from the latching state to the unlatching state by moving from the initial position to the actuation position, wherein an inner peripheral surface of the concave portion includes an inner peripheral surface formation portion that is located on a vehicle inner side of a vehicle-outer-side surface of the vehicle door and that is located at an end portion of the concave portion on one side in a front-rear direction of the vehicle door, the inner peripheral surface formation portion facing to a side substantially opposite to the one side in the front-rear direction of the vehicle door, and the latch release lever forms at least a part of the inner peripheral surface formation portion.
 2. The vehicle door handle device according to claim 1, wherein the inner peripheral surface formation portion is located at a rear end portion of the concave portion, and faces to a substantially front side.
 3. The vehicle door handle device according to claim 1, further comprising a key cylinder linked to the door lock device in such a way as to be able to switch the door lock device between a locking state of not allowing the door lock device to be switched from the latching state to the unlatching state and an unlocking state of allowing the door lock device to be switched from the latching state to the unlatching state, wherein the latch release lever and the key cylinder overlap with each other when viewed in an up-down direction.
 4. The vehicle door handle device according to claim 1, further comprising: a bell crank that is arranged to be movable between an initial position and an actuation position relative to the casing, and that is linked to the latch release lever in such a way that, when the latch release lever is located at the initial position, the bell crank is located at the initial position, and when the latch release lever is located at the actuation position, the bell crank is located at the actuation position; and a rod member that is linked to the bell crank and the door lock device in such a way as to switch the door lock device from the latching state to the unlatching state when the bell crank moves from the initial position to the actuation position.
 5. The vehicle door handle device according to claim 4, wherein the bell crank is configured in such a way as to move the rod member to a lower side by moving from the initial position to the actuation position.
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